Road Object Detection Under Adverse Weather Conditions

CSE437 Project Report | Group 4

Sayed Arifin Ahmed Alak (ID: 24241072) Farhan Mahmud (ID: 24241073)

Ramim Mahabub (ID: 23101480) Md Shahriar Saif (ID: 22299360)

Goal of the Project

The goal of this project is to develop and compare YOLOv8-based object detectors for recognizing road users and vehicles under adverse weather conditions using the DAWN dataset. The system focuses on six target classes - Person, Bicycle, Car, Motorcycle, Bus, and Truck - across Fog, Rain, Snow, and Sandstorm scenes. Because the dataset is severely imbalanced, the project emphasizes Precision, Recall, F1-score, and Average Precision from the Precision-Recall curve, while also reporting standard object-detection metrics such as mAP@50 and mAP@50-95. The final objective is to identify the YOLOv8 variant that gives the best overall test performance.

Materials and Method

Dataset: DAWN (Detection in Adverse Weather Nature), 1,027 labeled road images. Kaggle dataset link

- Weather distribution: Fog 300, Rain 200, Snow 204, Sandstorm 323 images.
- After cleaning, 7,845 target bounding boxes remained. Cars dominate with 6,454 boxes (82.27%), while Bicycle 26, Motorcycle 81, and Bus 161 are rare; therefore the dataset falls under the severe-imbalance group.

Data Preprocessing

- Inspected all annotation files, removed the single unsupported class-7 box, and kept only the six project target classes.
- Remapped raw sparse class IDs 1,2,3,4,6,8 to contiguous YOLO IDs 0-5. The one empty label (haze-012) was retained as a valid negative/background image.
- Used a fixed optimized split with seed 306: 718 train images, 154 validation images, and 155 untouched test images; no image leakage was found.

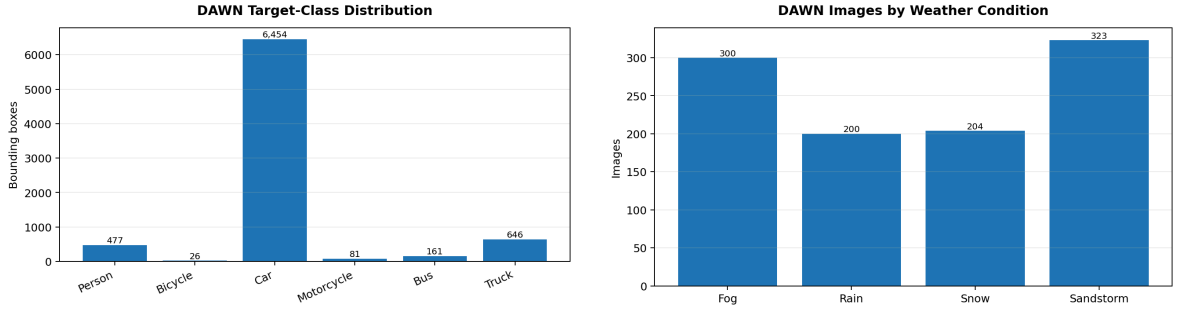


Figure 1. DAWN target-class distribution (left) and weather distribution (right).

ML Architectures and Training Setup

Three YOLOv8 variants were trained as a controlled comparison: YOLOv8n (Nano), YOLOv8s (Small), and YOLOv8m (Medium). All models used pretrained COCO weights, 640x640 input images, batch size 16, a maximum of 50 epochs, early stopping with patience 15, seed 306, and a Tesla T4 GPU. Keeping the split and core settings fixed allows the comparison to mainly reflect model-capacity differences.

Evaluation

For the severe-imbalance setting, the required metrics are Precision, Recall, F1-score, and AUPRC-style evaluation. In object detection, Average Precision (AP) summarizes the Precision-Recall curve and is used here as the object-detection counterpart of AUPRC. AP@50, mAP@50, and mAP@50-95 are also reported. The validation split was used during development, while the 155-image test split was kept untouched until final evaluation.

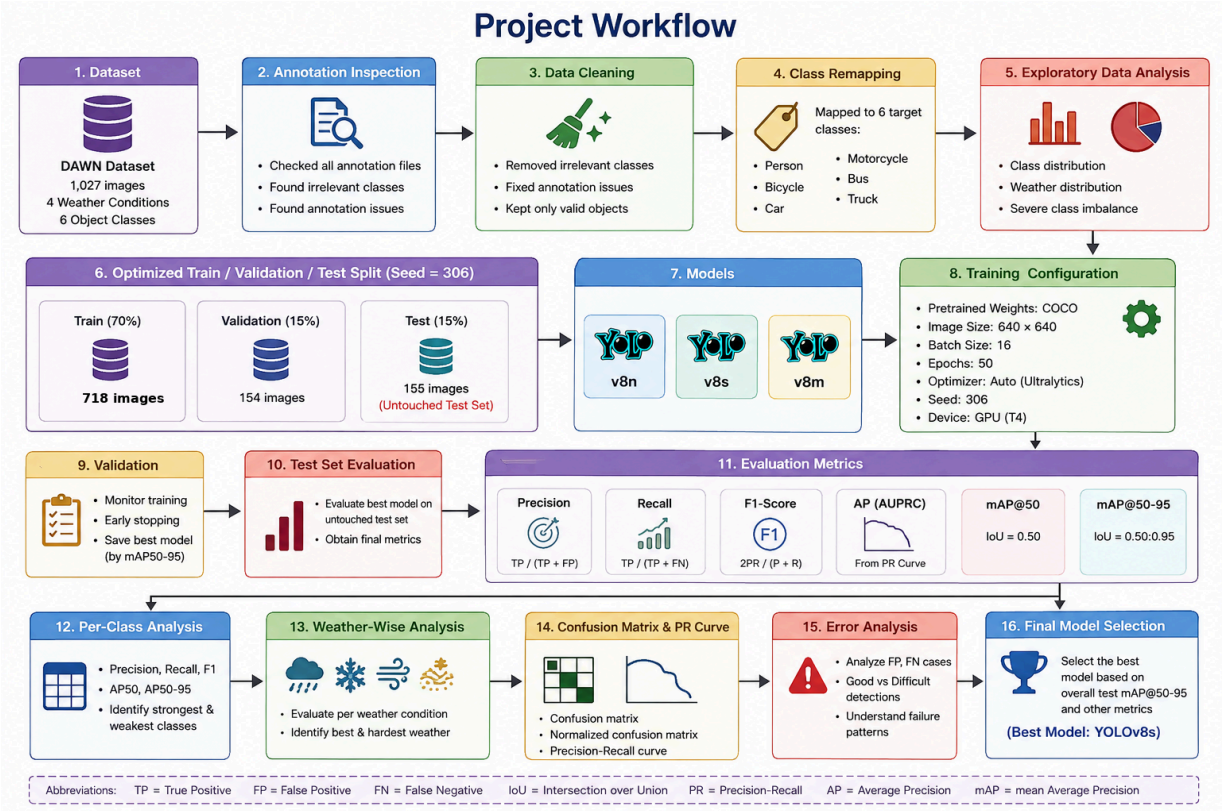


Figure 2. Full project workflow/pipeline from DAWN annotation inspection to final YOLOv8s selection.

Github Link: https://github.com/far-han12/CSE437-Project

Results

All final results below are from the untouched 155-image test set unless otherwise stated.

Model	Precision	Recall	F1	mAP@50	mAP@50-95
YOLOv8n	0.8279	0.5009	0.6242	0.5844	0.3552
YOLOv8s	**0.8728**	**0.5843**	**0.7000**	**0.6927**	**0.4253**
YOLOv8m	0.7242	0.6146	0.6649	0.6446	0.4069

Table 1. Final test-set comparison of the three trained YOLOv8 variants.

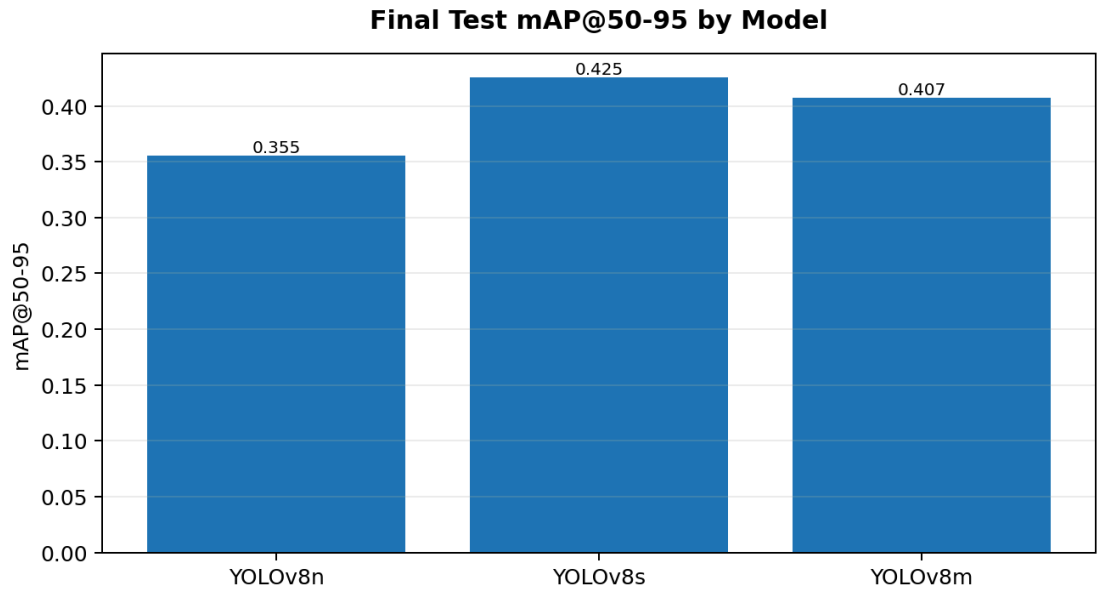


Figure 3. Final test mAP@50-95 by model.

YOLOv8s achieved the best overall result: Precision 0.8728, F1-score 0.7000, mAP@50 0.6927, and mAP@50-95 0.4253. YOLOv8m had the highest Recall (0.6146) but lower Precision and lower mAP@50-95. Therefore, YOLOv8s was selected as the final model. Compared with YOLOv8n, YOLOv8s improved mAP@50-95 by 0.0701 (7.01 percentage points), showing that increasing capacity from Nano to Small helped, while increasing further to Medium did not improve the main test metric.

Class-wise and Weather-wise Analysis

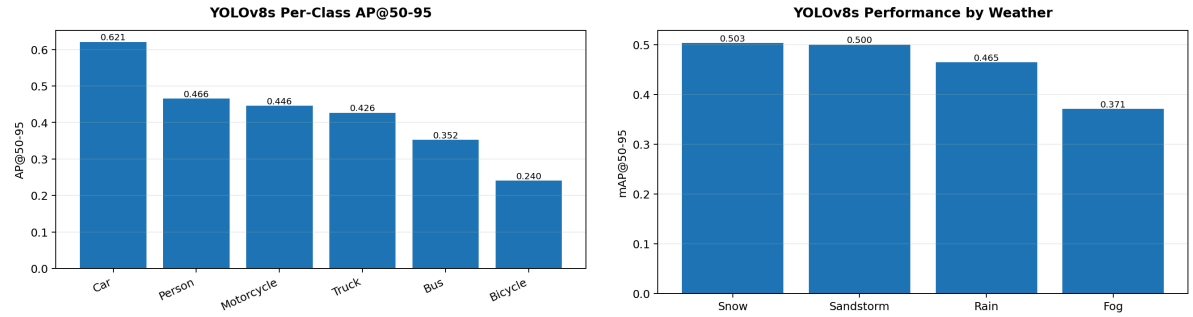


Figure 4. YOLOv8s per-class AP@50-95 (left) and weather-wise mAP@50-95 (right).

Strongest class **Car** AP@50-95 = 0.6207	**Weakest class** **Bicycle** AP@50-95 = 0.2403
Best observed weather **Snow** mAP@50-95 = 0.5032	**Hardest observed weather** **Fog** mAP@50-95 = 0.3708

The class-level results reflect the dataset imbalance. Car is the strongest class because it has far more training examples, while Bicycle is the weakest. Only 26 Bicycle boxes exist in the full dataset and only 4 are in the test split, so Bicycle performance is statistically unstable and should be interpreted cautiously. Weather-wise, Snow produced the highest observed mAP@50-95 and Fog the lowest. This is an observed result for the DAWN test split, not a universal claim about all foggy scenes, because weather subsets differ in object count and class composition.

Error Patterns

Qualitative inspection showed that the hardest images were dominated by haze/fog/mist scenes. The main errors were missed objects (false negatives) and occasional extra detections (false positives), especially when objects were small, partially obscured, or low-contrast. These errors are consistent with the lower Fog performance and the weaker rare-class results.

Visual Diagnostics

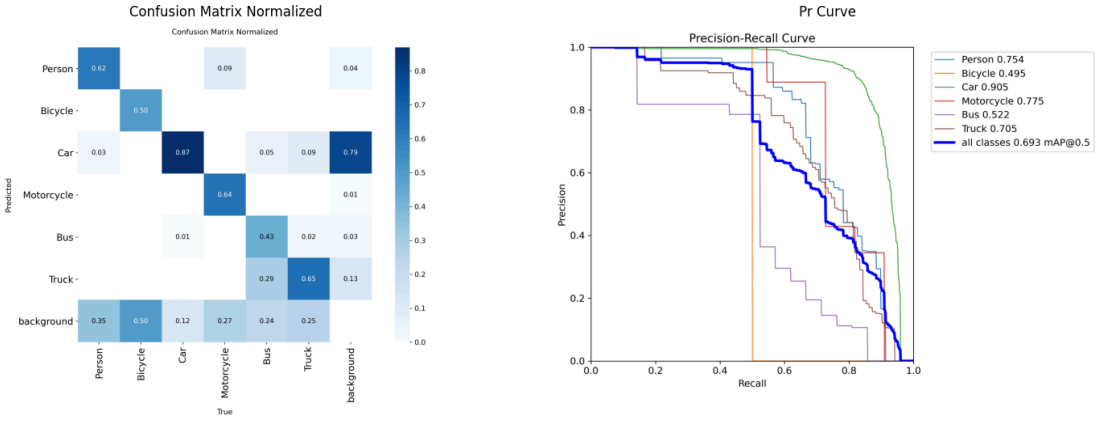


Figure 5. YOLOv8s normalized confusion matrix (left) and Precision-Recall curve (right).

The normalized confusion matrix shows the strongest diagonal response for Car (approximately 0.87), while rare classes have more background/missed-object errors. The Precision-Recall curve gives AP@50 values of approximately 0.754 (Person), 0.495 (Bicycle), 0.905 (Car), 0.775 (Motorcycle), 0.522 (Bus), and 0.705 (Truck), with an overall mAP@50 of about 0.693. These diagnostics support the same conclusion as the main test table: YOLOv8s performs strongly overall but remains limited by severe class imbalance and difficult low-visibility scenes.

Result Discussion

Across the three trained ML models, YOLOv8s provided the best overall balance of detection quality and consistency. It outperformed YOLOv8n substantially and also exceeded YOLOv8m on the main mAP@50-95 metric, despite YOLOv8m obtaining a slightly higher Recall. The final outcome therefore demonstrates that a larger model is not automatically better for this dataset. The main remaining limitations are the severe class imbalance, the very small Bicycle sample, and the difficulty of fog/haze scenes. Future improvement should focus on collecting more rare-class examples and applying weather- and minority-class-aware augmentation.